

Chapter 16.1

AutoEncoders

(Unsupervised Learning + Self-Supervised)

தமிழில்

Less than 60 Mins

AI COACH JOHN

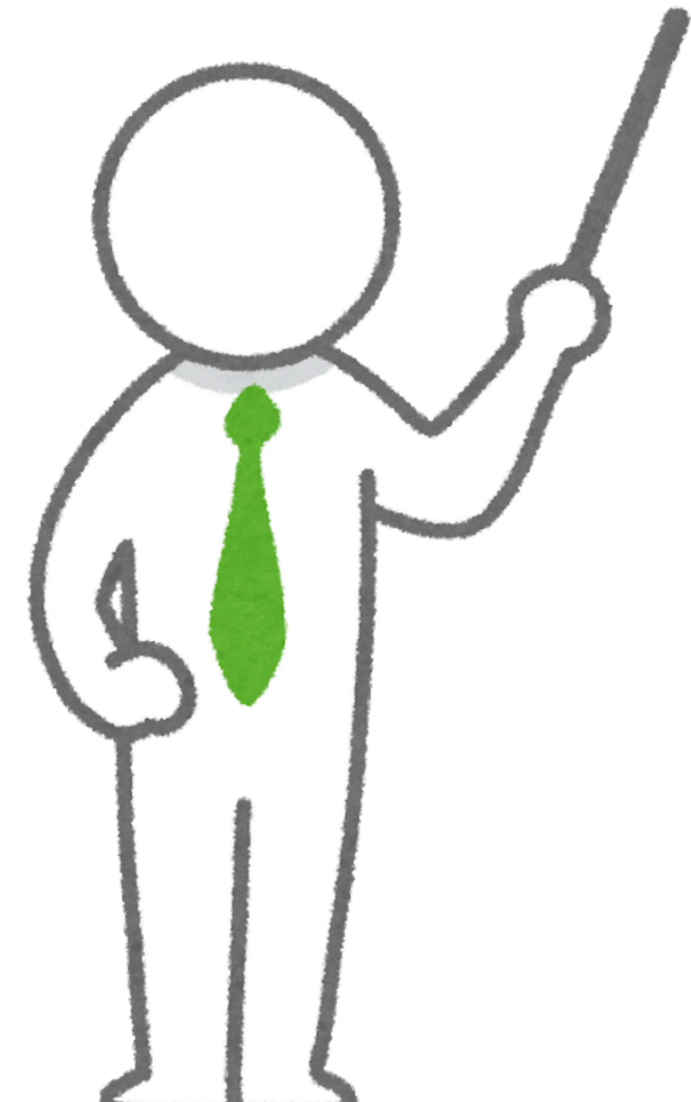
CEO & FOUNDER

Agenda at a glance

1. **Technical Assessment Question**
2. How to approach solving a Problem?
3. **Solution** → Step-by-Step Implementation
4. What is AutoEncoders?
5. Takeaway

Point 1

Technical Assessment - 1





Do the RECONSTRUCTION



Point 2

How to approach solving a Problem?

How to approach solving a Problem?

Step 1: Read and Understand the **Business Problem**

Step 2: Study the **whereabouts** of the company

Step 3: Read the complete document again

Step 4: Connect the tools you've learned and start thinking in **systems** → not just code.

Step 5: Read how to submit the Assessment and the TAT very clearly

Step 6: Submit it ON-Time



Never Expect, Keep Hunting....

Point 3

Solution → *Step-by-Step Implementation*

AutoEncoders – Step By Step Approach

Step 1: Import Libraries

Step 2: Import Dataset

Step 3: Data Preparation/Pre-processing
(*Normalize → add noise → clip*)

Step 4: Model Building (Encoder/Decoder)

Point 4

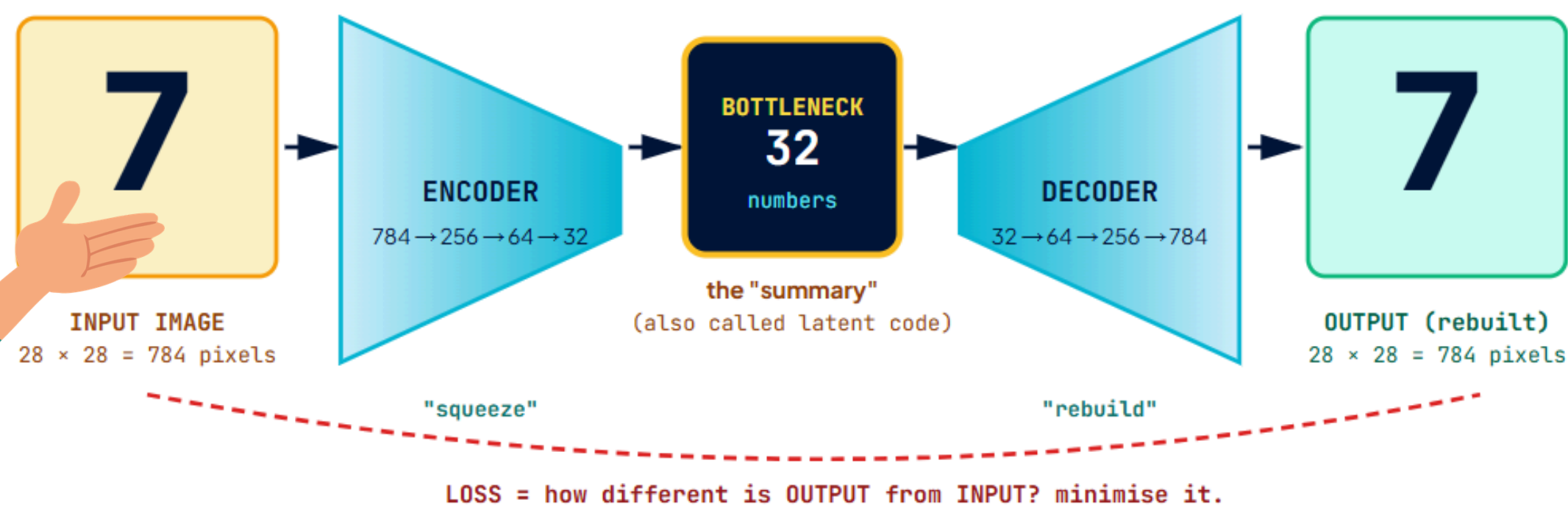
*What is an **AutoEncoder**?*

Point 4 What is an AutoEncoder?

An autoencoder is just **2 Neural Networks** *glued* together

- The first half (encoder) **compresses**;
- The second half (decoder) **reconstructs**.
- They meet at the narrowest point → **The bottleneck** → **The Latent Space**

where the compressed summary lives.



AutoEncoders – Step By Step Approach

Step 1: Import Libraries

Step 2: Import Dataset

Step 3: Data Preparation/Pre-processing
(*Normalize → add noise → clip*)

Step 4: Model Building (Encoder/Decoder)

Step 5: Model Compilation

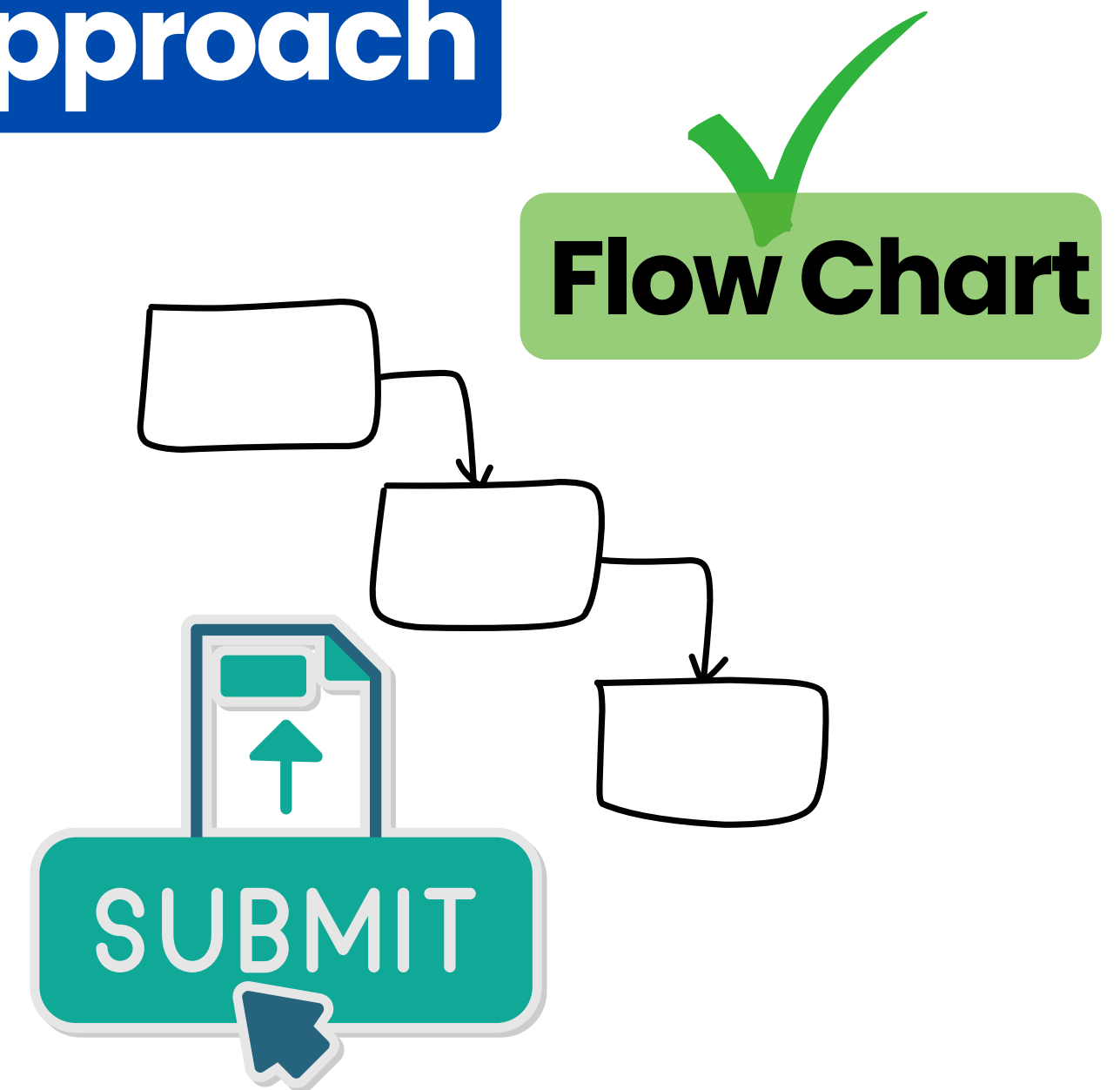
Step 6: Model Training (*fit*)

Step 7: Evaluate Model

Step 8: Test on Sample Image

Step 9: Save Trained Model

Step 10: Deploy via No-Code Platform

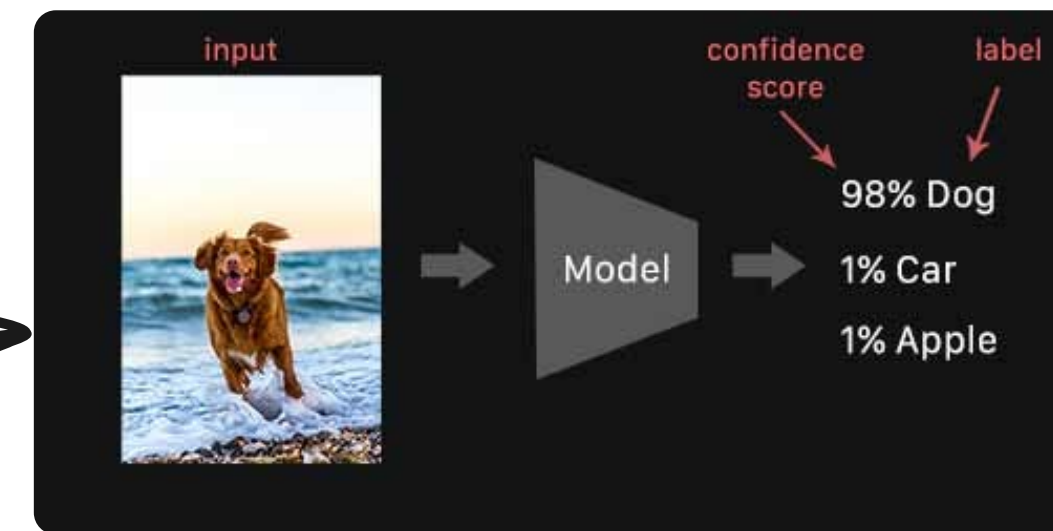


Point 5

Takeaway

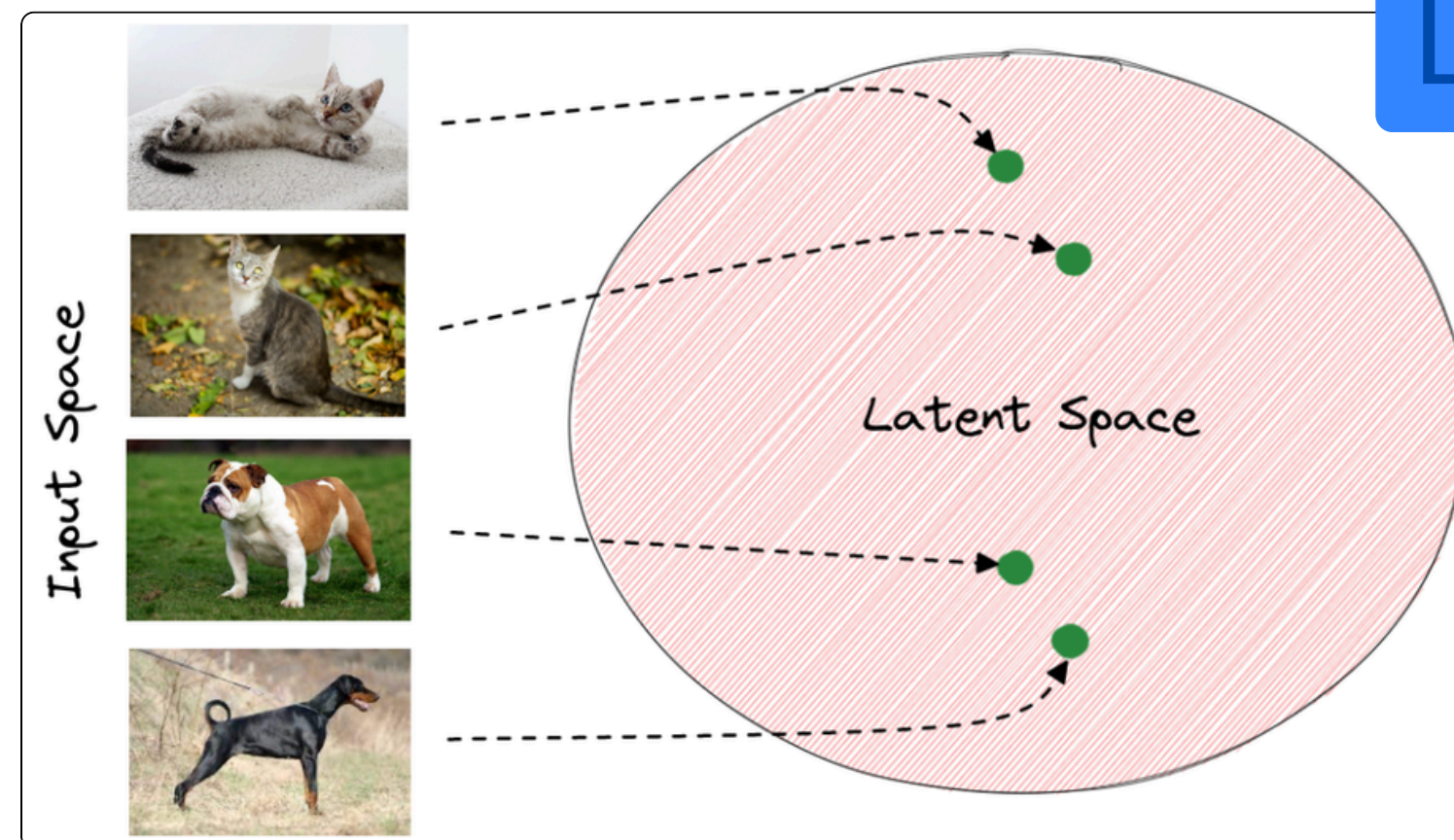
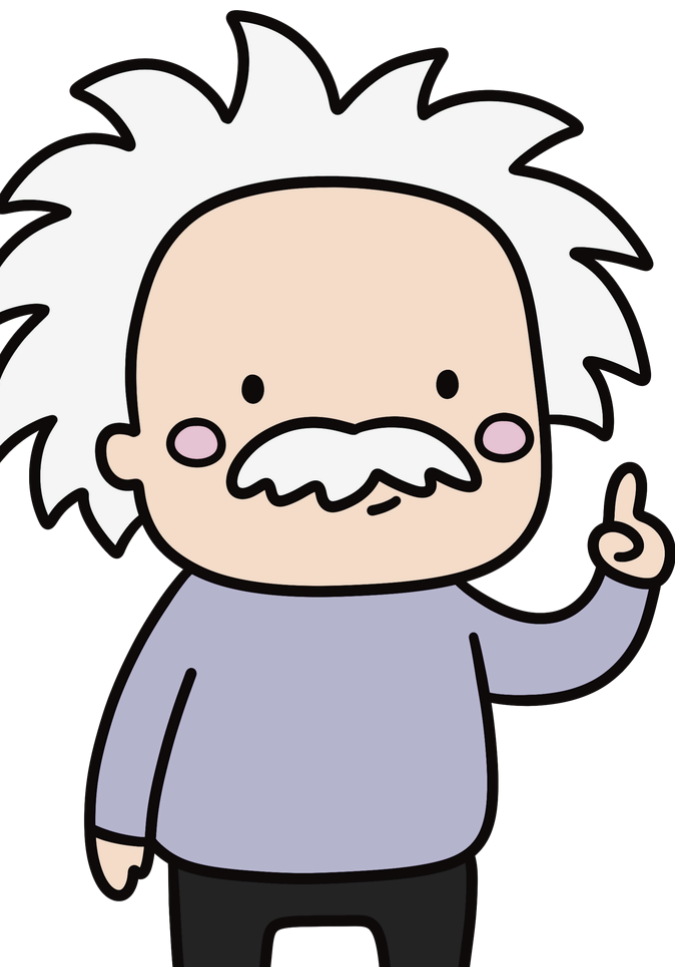
Key Takeaway Point

- **CNN Era:** Input → Prediction



- **Autoencoder Era:** Input → Meaningful Representation

“Instead of learning how to classify data...
the network **started learning** how to represent data itself.”



Latent Space

Summary

1. **Technical Assessment Question**
2. How to approach solving a Problem?
3. **Solution** → Step-by-Step Implementation
4. What is AutoEncoders?
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